

Homework Classical Physics, Fall 2023

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Problem Set 01.1 (L01 & L02)

Due on Thursday, 14. May 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - units and geometry (4 points)

(a) A car needs 34.7 liter gasoline for a drive of 647 km. How much fuel does it need for 100 km? In the US people will ask you how far your car can get with a filling of one gallon of fuel. What is the consumption of the car in miles per gallon? (1 US gallon = 3785.4 cm³, 1 mile = 1609.3 m)

(b) You are at the coast and your eyes are 3.0 m above sea level. What is the farthest distance from the coast a ship can be, so that you just can see the top light of the ship which is 20 m above the water level? (assume radius of Earth of about 6370 km)